PRANA-G 100M: Checkpoint Analysis Report

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Checkpoint `ckpt_best.pt` (step 8,250, 100.1M parameters, 12 layers, hidden size 768) from the 100M training run on an A100 MIG slice, trained on 17 domains. All 34 training and validation files used here were checked against the run's preflight log and match it token for token. Every number below comes from the result files in `results/`; §8 gives the one command that reproduces them.

1 Training and validation loss

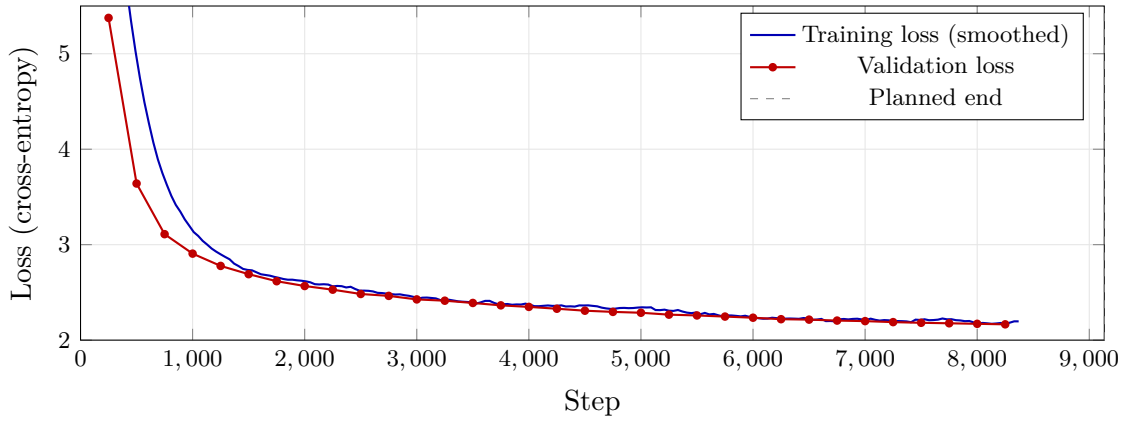


Figure 1: Training loss (smoothed) and validation loss on the mixed validation set, every 250 steps.

Training loss, first logged step / end (smoothed)	10.14 / 2.199
Validation loss, first evaluation (step 250) / best	5.38 / 2.166 at step 8,250 (perplexity 8.72)
Training minus validation loss at the end	+0.001
Validation loss change over the last 1,000 steps	-0.023 per 1,000 steps
Steps completed	8,380 of 9,134 planned (91.7%), 2.20B tokens
Gradient norm (median / max)	0.23 / 2.60; 35 spikes above 3× median
Throughput, peak memory	33,458 tokens/s, 9.2 GB

Table 1: Loss summary (from `metrics.jsonl`).

- **No overfitting.** Training and validation loss are equal at the end (difference +0.001), so the model is not memorising its data.
- **Not converged.** Validation loss was still falling (-0.023 per 1,000 steps) when the run stopped; more tokens would still help.
- **The run did not finish.** It stopped at step 8,380 of 9,134 when the GPU slot ended. The last checkpoint was saved at step 8,250, so the final 130 steps were lost.
- **Training was stable.** All losses are finite. Almost all gradient-norm spikes happened in the first 400 steps (warm-up); after step 1,000 there were only two small ones (1.76 at step 1,810 and 0.95 at step 4,180).

2 Domain-wise perplexity and next-token prediction

Each domain's validation file was cut into the same non-overlapping 2,048-token windows (100 windows, about 200k tokens; astronomy and finance have fewer). For every position the model predicts the next token: *cross-entropy* is the average loss, *perplexity* is e raised to the cross-entropy, *top-1 / top-5* is how often the true token is the model's first choice / among its five most likely, and *calibration error* is the gap between how confident the model is and how often it is right.

Domain	Tokens	Cross-entropy	Perplexity	Top-1	Top-5	Calibration error
cad	204,800	0.217	1.24	94.7%	98.2%	0.2%
code	204,800	1.276	3.58	74.0%	87.6%	0.8%
bio	204,800	1.306	3.69	73.7%	85.1%	0.8%
finance	86,016	1.363	3.91	73.8%	81.7%	4.7%
engineering	204,800	1.440	4.22	70.3%	84.6%	0.5%
scidisc	204,800	1.566	4.79	67.6%	81.4%	0.4%
sciprog	204,800	1.732	5.65	65.6%	82.6%	1.2%
multistep	204,800	1.883	6.58	59.8%	75.7%	0.8%
earth	204,800	1.938	6.95	63.2%	77.4%	0.3%
gen_eng	204,800	2.349	10.47	53.0%	73.7%	1.2%
sciqa	204,800	2.432	11.38	50.8%	72.7%	0.9%
chem	204,800	2.445	11.53	50.3%	72.1%	0.6%
exp_val	204,800	2.635	13.94	49.4%	69.4%	0.7%
physics	204,800	2.750	15.65	46.0%	67.5%	0.3%
math	204,800	2.773	16.00	47.2%	67.8%	0.8%
astro	100,352	2.983	19.74	42.9%	64.8%	0.6%
tech_comm	204,800	3.108	22.38	43.9%	63.0%	0.6%

Table 2: Next-token prediction for all 17 training domains, sorted by perplexity.

- **Very low perplexity points to templated data.** CAD (perplexity 1.24, top-1 95%) is almost fully predictable, and code, biology and finance (3.6–3.9) are also unusually easy. Text this predictable is usually generated from templates or highly repetitive; such data teaches the model little and should be checked for diversity before the 30B run.
- **The hardest domains are technical communication, astronomy, maths and physics** (perplexity 16–22, top-1 43–47%), which are also the most varied natural text.
- Multi-step reasoning is easy at the token level (perplexity 6.6) but the model solves almost no problems (§5): predicting the style of a worked solution is not the same as solving it.
- Calibration is good: below 1.5% error in every domain except finance (4.7%), so the model's confidence matches how often it is right.

3 Generation quality, repetition and consistency

Probes. Ten fixed one-line prompts with an automatic check, answered greedily and with three seeded samples (temperature 0.8).

Domain	Prompt	Greedy	Samples passing	Loop
arithmetic	`Question: What is 17 * 24? Answer:`	fail	0/3	no
arithmetic	`Question: What is 125 + 379? Answer:`	fail	0/3	no
math	`The derivative of x^2 with respect to x is`	pass	1/3	no
code	`def add(a, b): return`	pass	2/3	no
code	`def factorial(n): if n == 0: ret`	fail	0/3	yes
chemistry	`The chemical formula of water is`	fail	0/3	yes
physics	`Newton's second law of motion states that`	pass	1/3	no
physics	`The speed of light in vacuum is approximatel`	fail	0/3	yes
biology	`DNA is made of four nucleotides:`	fail	1/3	yes
earth	`The three main types of rock are`	fail	0/3	no

Table 3: Probe results. "Loop" means the greedy answer keeps repeating itself.

The automatic check passes 3 of 10 greedy answers, but reading them, only `def add(a, b): return a + b` is really correct: the physics answer matched the word "force" in a wrong statement, and the maths answer matched "2x" inside unrelated algebra. Wrong facts are stated confidently ("the speed of light in vacuum is approximately 1.5 times the speed of light").

Repetition: greedy decoding against sampling. Six held-out documents per domain were cut after 64 tokens and continued for 96 tokens, once greedily and once with a seeded sample (temperature 0.8). The table shows the share of repeated 4-grams and the number of continuations that fall into a loop.

Domain	Texts	Real text	Greedy	Greedy loops	Sampled (T=0.8)	Sampled loops
math	6	5%	64%	6	5%	0
physics	6	1%	32%	1	5%	0
chem	6	2%	68%	6	16%	0
bio	6	5%	18%	1	4%	0
earth	6	1%	30%	2	6%	0
scidisc	6	2%	53%	4	9%	0
sciqa	6	5%	40%	3	12%	0
sciprog	6	16%	70%	6	23%	2
code	6	16%	55%	4	22%	2
cad	6	16%	15%	0	16%	0
multistep	6	0%	0%	0	0%	0
astro	6	2%	49%	4	2%	0
finance	6	8%	41%	4	9%	0
engineering	6	0%	6%	0	1%	0
gen_eng	6	4%	57%	4	4%	0
exp_val	6	1%	63%	5	7%	0
tech_comm	6	1%	72%	6	1%	0
All	102			56		4

Table 4: Repeated 4-grams in the real continuation, in greedy output and in sampled output, and repetition loops.

Most of the looping comes from greedy decoding. Greedy continuations loop in 56 of 102 texts; sampled continuations of the same prompts loop in only 4, and repeat about as much as real text. Greedy decoding always takes the single most likely token, and a small model's most likely continuation is often the text it has just written. Evaluation and demos should therefore use sampling (or a repetition penalty), and SFT must still teach the model to stop.

Consistency.

- Greedy decoding gives identical text when repeated (10/10 probes).

- The three samples agree with each other on 6 of 10 probes, mostly because they all fail; correct answers are not stable across samples.
- The two checkpoints 50 steps apart give the same benchmark answer on 91.7% of questions, and their overall accuracy differs by 0.06 points (26.9% and 26.9%).
- Scorer check: reversing the order of the options changed none of 400 multiple-choice answers, so the benchmark cannot favour a position.

4 Checkpoint integrity and reproducibility

Check	`ckpt_best.pt`	`ckpt_last.pt`
File size	1.20 GB	1.20 GB
SHA-256 (first 16)	`54252dfec6573c82`	`2b72125a3bdb2caf`
Step	8,250	8,200
Parameters	100,092,672	100,092,672
Tensors	110	110
All weights finite	yes	yes
Tokenizer hash matches	yes	yes
Optimizer state saved	yes	yes
Validation loss stored	2.1656	–
Matches metrics.jsonl	yes	–
Saved with PyTorch	2.10.0+cu128	2.10.0+cu128

Table 5: Integrity of the two checkpoints.

- Both files load, all weights are finite, the tokenizer hash matches, and the validation loss stored in `ckpt_best.pt` equals the value in the training log exactly.
- **Why the validation loss was reported as null.** Validation loss is measured every 250 steps, but checkpoints are saved every 200 steps. `ckpt_last.pt` is from step 8,200, which is not an evaluation step, so it has no validation loss stored, and the sample-output script printed it as null. `ckpt_best.pt` (step 8,250) has it: 2.166. Nothing is missing from the training itself. Note that `ckpt_last.pt` is 50 steps *older* than `ckpt_best.pt`, because the run stopped after step 8,250's save.
- The two checkpoints have the same configuration and tensors and differ by 1.17% of their weights (50 steps of training).
- Repeating a forward pass gives identical outputs (max difference 0); CPU and Apple-GPU outputs differ by at most 2.2e-05 and predict the same token at every position (100.0%).
- The benchmark re-scores its first 20 questions on every run and requires identical answers; this passed on every run. The whole analysis was produced by one command (§8).

5 Re-evaluation of the benchmark

The benchmark is frozen at version 2.1: 3,619 questions in 19 domains (the first set of 2,340 questions, of which 2,337 were scored in the first report, plus biology, astronomy, finance, earth science, scientific discovery, scientific programming and CAD), with a SHA-256 hash for every file.

Leakage. Every question was searched for in all 17 training files. 75 were found and removed from scoring, leaving 3,544. 39 are in `sciqa.train.bin`, which contains maths questions with worked answers that include MMLU high-school maths questions word for word (median match 95%); 28 are in `math.train.bin` (forum copies), and 8 in physics, technical communication and chemistry. Maths is hit hardest: 55 of its 400 questions were removed. The `sciqa` text also carries leftover source tags ("GSM_AnsAug", "Expected Answer:"), like the multistep data.

Results. Overall accuracy is 26.9% against 24.8% for random guessing.

Domain	n	Accuracy	95% CI	Chance	Above chance
astro	150	32.7%	25.7%–40.5%	25.0%	yes
bio	290	25.5%	20.8%–30.8%	25.0%	no
cad	1	0.0%	0.0%–79.3%	25.0%	no
chemistry	252	25.8%	20.8%–31.5%	24.5%	no
code	202	24.8%	19.3%–31.1%	25.0%	no
earth	274	32.1%	26.9%–37.9%	25.0%	yes
electrical_eng	143	32.2%	25.1%–40.2%	25.0%	yes
engineering	40	12.5%	5.5%–26.1%	18.8%	no
exp_val	24	20.8%	9.2%–40.5%	19.8%	no
finance	358	24.6%	20.4%–29.3%	25.0%	no
math	345	22.3%	18.2%–27.0%	25.0%	no
multistep	200	1.5%	0.5%–4.3%	0.0%	yes
physics	409	25.9%	21.9%–30.4%	24.7%	no
scidisc	200	50.0%	43.1%–56.9%	50.0%	no
sciml	16	25.0%	10.2%–49.5%	25.0%	no
sciprog	4	25.0%	4.6%–69.9%	25.0%	no
sciqa	599	30.4%	26.8%–34.2%	25.0%	yes
simulation	22	31.8%	16.4%–52.7%	25.0%	no
tech_comm	15	33.3%	15.2%–58.3%	25.0%	no

Table 6: Benchmark accuracy by domain (multiple choice: random guessing 25%, PubMedQA yes/no 50%; GSM8K numeric: 0%).

- **Above chance:** school science (sciqa 30.4%), earth science (32.1%), astronomy (32.7%) and electrical engineering (32.2%). Multistep is formally above chance only because its chance level is 0 (3 of 200 solved).
- **At chance:** maths, physics, chemistry, biology, finance, code and every hand-written PRANA-G domain.
- **Scientific discovery:** the model answers "yes" to all 200 PubMedQA questions whatever the abstract says (50%).

6 Scientific capability gap analysis

The table maps the 20 domains of the 30B token plan to what the 100M run trained on, its benchmark result and its validation perplexity.

Domain (30B plan)	Plan	100M run	Benchmark accuracy	Val. ppl	Status
Mathematics	2.4B	0.395B	math 22% (ch. 25%, n=345)	16.0	at chance
Physics	2.1B	0.209B	physics 26% (ch. 25%, n=409)	15.6	at chance
Chemistry	1.2B	0.070B	chemistry 26% (ch. 25%, n=252)	11.5	at chance
Biology	1.2B	0.047B	bio 26% (ch. 25%, n=290)	3.7	at chance
Earth & Environment	0.9B	0.070B	earth 32% (ch. 25%, n=274)	6.9	above chance
Scientific Discovery	1.2B	0.279B	scidisc 50% (ch. 50%, n=200)	4.8	at chance
Scientific Q&A	1.5B	0.116B	sciqa 30% (ch. 25%, n=599)	11.4	above chance
Scientific Programming	1.5B	0.047B	sciprog 25% (ch. 25%, n=4)	5.7	at chance (few questions)
Code	1.8B	0.233B	code 25% (ch. 25%, n=202)	3.6	at chance
CAD/STEP	1.2B	0.023B	cad 0% (ch. 25%, n=1)	1.2	at chance (few questions)
Multi-step Reasoning	2.1B	0.023B	multistep 2% (ch. 0%, n=200)	6.6	above chance
Astronomy	0.9B	0.012B	astro 33% (ch. 25%, n=150)	19.7	above chance
Finance	0.6B	0.012B	finance 25% (ch. 25%, n=358)	3.9	at chance
Engineering Agent/Tool Use	1.2B	0.163B	–	4.2	no benchmark questions
Engineering Reasoning	2.4B	0.000B	electrical_eng 32% (ch. 25%, n=143); engineering 12% (ch. 19%, n=40)	–	not in training data
Generative Engineering	1.8B	0.163B	–	10.5	no benchmark questions
Experimental Validation	1.2B	0.163B	exp_val 21% (ch. 20%, n=24)	13.9	at chance (few questions)
Technical Communication	0.6B	0.140B	tech_comm 33% (ch. 25%, n=15)	22.4	at chance (few questions)
Numerical Simulation	2.1B	0.000B	simulation 32% (ch. 25%, n=22)	–	not in training data (few questions)
Physics + AI (PINNs/FNO)	2.1B	0.000B	sciml 25% (ch. 25%, n=16)	–	not in training data (few questions)

Table 7: Capability gaps. "Plan" is the 30B allocation; "100M run" is the approximate number of tokens of that domain the model saw.

1. **Three plan domains had no training data:** Engineering Reasoning (2.4B planned; its file was rebuilt but left out of the run), Numerical Simulation and Physics + AI (2.1B each, no data yet). Their benchmark scores are at chance.
2. **University-level knowledge is absent** in every domain; only school-level science and electrical engineering are above chance.
3. **Reasoning and reading are absent:** 3 of 200 word problems solved; on PubMedQA the model answers "yes" to everything.
4. **Generation is not usable with greedy decoding** (loops) and states wrong facts confidently.
5. **Data quality limits the run:** benchmark copies in sciqa and maths, leftover source tags in sciqa and multistep, and near-templated text in CAD, finance, code and biology.
6. **The benchmark is thin where PRANA-G matters most:** the hand-written domains have 1–40 questions each, and engineering agent/tool use (7% of this run's training) has no test.

7 SFT preparation and next training stage

SFT teaches a model to use what it already knows in a given format; it cannot add knowledge that pretraining did not give it. At 100M parameters and 2.2B tokens most failures are knowledge failures, so the list starts with what must be fixed before SFT is worth running.

Priority 1: before SFT (next pretraining stage).

1. **Train on all 20 domains of the 30B plan**, including Engineering Reasoning (left out of this run), Numerical Simulation and Physics + AI (no data yet): 6.6B of the 30B tokens.
2. **Clean and decontaminate the data:** remove the 75 benchmark questions found in sciqa, maths, physics, technical communication and chemistry; strip the leftover source tags ("Benchmark source: data/CoT/…" in multistep, "GSM_AnsAug" and "Expected Answer:" in sciqa); check CAD, finance, code and biology for templated or repeated text.

3. **Make runs finish and keep their result:** save a checkpoint at the last step and copy checkpoints off the workspace before it expires (this run lost its last 130 steps); save checkpoints on evaluation steps so every saved checkpoint has a validation loss.
4. **Track capability during training:** run the benchmark and the per-domain perplexity on every saved checkpoint, next to the loss curve.

Priority 2: SFT data, in order of the failures observed.

1. **Answer format and stopping** (loops; only 68 of 200 GSM8K answers reach a final answer line): short question–answer pairs that end with an explicit final answer and an end-of-text token.
2. **Worked multi-step solutions** (1.5% on GSM8K, arithmetic probes wrong): step-by-step solutions with checked final answers in maths, physics and engineering, with units.
3. **Answering from a given text** (always "yes" on PubMedQA): context–question–answer examples with balanced yes/no answers and examples where the text does not contain the answer.
4. **Saying "cannot be determined"** when information is missing: examples where the correct response names the missing quantity.
5. **PRANA-G tasks:** engineering calculations with units, simulation set-up and verification, PINN and neural-operator explanations, validation reports and technical writing, once pretraining covers these domains.

Priority 3: evaluation for the next stage.

1. Use sampling (or a repetition penalty) for all generation-based evaluation, and report greedy and sampled results side by side.
2. Grow each hand-written domain to at least 50 questions; add an executable code test for sciprog, CAD/STEP questions and an agent/tool-use test.
3. Score the 1.5B checkpoints on exactly this frozen benchmark, so the 100M and 1.5B results can be compared directly.

8 How to reproduce

Code: branch `namit/science-eval` of `github.com/Pankaj837/LLM`. Inputs: the checkpoint zip from the training run and the 34 `.bin` files from the team Drive (the same files as the run's `dl.sh`); their SHA-256 hashes are in `results/inputs_sha256.txt`.

```
RUN=path/to/checkpoint_folder BINS=path/to/bins bash scripts/reproduce_analysis.sh
```

This records the input hashes, runs the leakage check, the benchmark on both checkpoints and the analysis, regenerates every table and builds this PDF. All steps are deterministic (fixed windows, greedy decoding, seeded sampling); the tests in `tests/` cover each script.